Es1 $W \subsetneq \mathbb{R}^5$ definito da $\begin{cases} x_1 - x_3 + x_4 = 0 \\ x_1 + x_2 + x_5 = 0 \\ x_3 + x_4 + x_5 = 0 \end{cases}$ $A.X = \vec{0}$

$$A = \begin{bmatrix} 1 & 0 & -1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

$c_1 c_2 c_3$

$$\begin{vmatrix} 1 & 0 & -1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \neq 0 \Rightarrow \rho(A) = 3$$

$$\dim W = n - \rho(A) = 5 - 3 = 2$$

$$A \xrightarrow{R_2 \to R_2 - R_1} \begin{bmatrix} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & -1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \qquad \begin{cases} x_1 - x_3 + x_4 = 0 \\ x_2 + x_3 - x_4 + x_5 = 0 \\ x_3 + x_4 + x_5 = 0 \end{cases}$$

Alg. Gauss

Triangolare x_4, x_5 inc. libere

$$\begin{cases} x_1 = -2x_4 - x_5 \\ x_2 = 2x_4 \\ x_3 = -x_4 - x_5 \end{cases}$$

$$W = \left\{ {}^t\left[-2x_4 - x_5,\ 2x_4,\ -x_4 - x_5,\ x_4,\ x_5\right] \in \mathbb{R}^5 : x_4, x_5 \in \mathbb{R} \right\}$$

$\mathcal{B} = \{\vec{w}_1, \vec{w}_2\}$

base di W

$\vec{w}_1 : x_4 = 0\ x_5 = 1\ ;\ \vec{w}_1 = \begin{bmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$

$\vec{w}_2 : x_4 = 1\ x_5 = 0\ \ \vec{w}_2 = \begin{bmatrix} -2 \\ 2 \\ -1 \\ 1 \\ 0 \end{bmatrix}$

base ortonormale $\mathcal{B}' = \{\vec{w}'_1, \vec{w}'_2\}$:

$$\vec{w}'_1 = \text{vers}(\vec{w}_1) = \frac{\vec{w}_1}{|\vec{w}_1|} = {}^t\left[-\frac{1}{\sqrt{3}}, 0, -\frac{1}{\sqrt{3}}, 0, \frac{1}{\sqrt{3}}\right]$$

$$\vec{w}'_2 = \text{vers}\left(\vec{w}_2 - (\vec{w}_2 \cdot \vec{w}'_1)\vec{w}'_1\right) \quad \text{calcoli...}$$

Suggerimento: verifica finale $\begin{cases} \vec{w}'_1 \cdot \vec{w}'_2 = 0 \\ \vec{w}'_1 \text{ e } \vec{w}'_2 \in W \end{cases}$

soddisfano il sistema che definisce W

Es2 $f : \mathbb{R}^3 \to \mathbb{R}^3$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \longmapsto \begin{bmatrix} x_1 + 2x_2 - 3x_3 \\ 0 \\ 0 \end{bmatrix} \qquad f(X) = A.X$$

$A = ?$

(a) $A = \left[f(\vec{e}_1), f(\vec{e}_2), f(\vec{e}_3)\right] = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ colonna 1

(b) $A = A_{sym} + A_{antisym}$

$$A_{sym} = \frac{A + {}^tA}{2} = \begin{bmatrix} 1 & 1 & -3/2 \\ 1 & 0 & 0 \\ -3/2 & 0 & 0 \end{bmatrix}$$

$$A_{antisym} = \frac{A - {}^tA}{2} = \begin{bmatrix} 0 & 1 & -3/2 \\ -1 & 0 & 0 \\ 3/2 & 0 & 0 \end{bmatrix}$$

(c) dim Ker f e base ortonormale di Ker f

Ker f è definito da $\underbrace{A.X = \vec{0}}$ $\quad A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\dim \text{Ker} f = n - \rho(A) = 3 - 1 = 2$

$$\Longleftrightarrow \begin{cases} x_1 + 2x_2 - 3x_3 = 0 \\ 0 = 0 \\ 0 = 0 \end{cases} \qquad \mathcal{B} = \left\{ \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$\mathcal{B}'$ --- ortonormalizzazione --- –

(d) $\dim(\text{Im} f)$ e base o.n. di Im f

$$\dim(\text{Im} f) = \rho(A) = 1 \qquad \mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

(e) A è diagonalizzabile?

$$P(\lambda) = |A - \lambda I| = \begin{vmatrix} (1-\lambda) & 2 & -3 \\ 0 & -\lambda & 0 \\ 0 & 0 & -\lambda \end{vmatrix} = \lambda^2(1-\lambda)$$

Tutte le radici di $P(\lambda)$ sono reali.

Abbiamo 2 autovalori $\begin{cases} \lambda_1 = 0 & m_a(\lambda_1) = \boxed{2 = m_g(\lambda_1)} \\ \lambda_2 = 1 & m_a(\lambda_2) = 1 = m_g(\lambda_2) \end{cases}$

verificare

$\Rightarrow$ $\boxed{\text{A è diagonalizzabile}}$

$\lambda_1 = 0$

$$m_g(\lambda_1) = n - \rho(A - \lambda_1 I) = 3 - \underbrace{\rho(A)}_{=1} = 2$$

Quindi $\exists$ P invertibile, $P \in M_3(\mathbb{R})$, t.c.

$$P^{-1}.A.P = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P = \left[\underbrace{\vec{w}_1, \vec{w}_2}_{\substack{\text{base di } V_{\lambda_1} \\ = \text{Ker} f}}, \underbrace{\boxed{\vec{w}_3}}_{V_{\lambda_2} \text{ base}}\right] = \begin{bmatrix} 3 & -2 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

verificare

$\vec{w}_3$ si ricava da $(A - \lambda_2 I)X = \vec{0}$

(f) $Tr(A) - Tr(P^{-1}AP) = 0$ Sappiamo che Tr

$\det A - \det(P^{-1}.A.P) = 0$ e det sono

2 invarianti di f, cioè non dipendono dalla base in cui sono calcolati. Quindi

$\gamma : \underbrace{x^2 + 8xy + y^2} - 5 = 0$ Caso $a_{12} \neq 0$

$$A = \begin{bmatrix} 1 & 4 \\ 4 & 1 \end{bmatrix} \rightsquigarrow P(\lambda) \overset{\text{calcolare}}{=} (\lambda - 5)(\lambda + 3) \qquad \begin{matrix} \lambda_1 = 5 \\ \lambda_2 = -3 \end{matrix}$$

$\exists$ rotazione P t.c. ${}^tP.A.P = \begin{bmatrix} 5 & 0 \\ 0 & -3 \end{bmatrix}$

Si trova $P = [\vec{w}_1, \vec{w}_2] = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$ dove $\begin{bmatrix} x \\ y \end{bmatrix} = P \begin{bmatrix} x' \\ y' \end{bmatrix}$

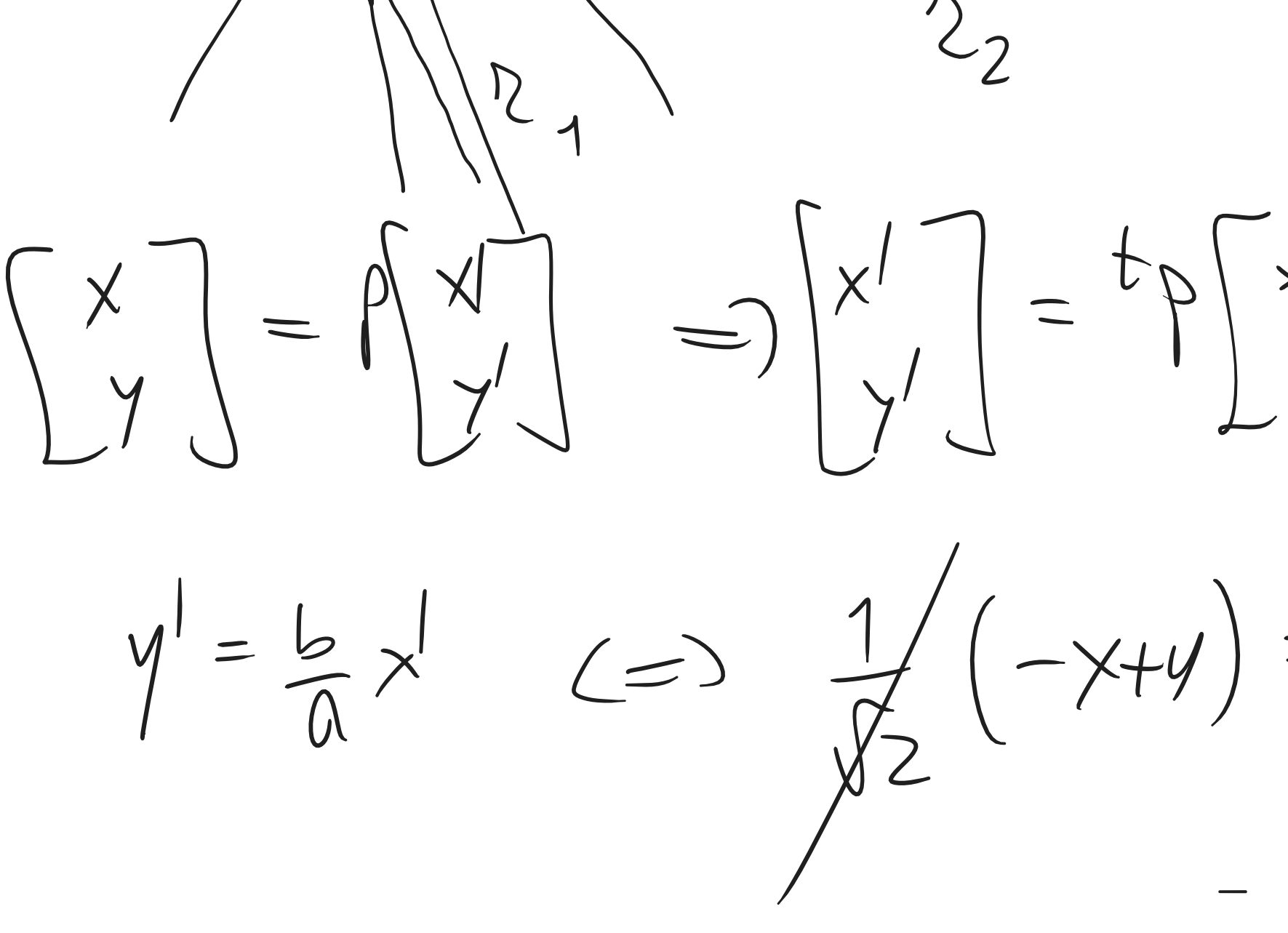
$\gamma : \lambda_1 x'^2 + \lambda_2 y'^2 - 5 = 0$

$\gamma : x'^2 - \dfrac{y'^2}{\left(\frac{5}{3}\right)} = 1$ iperbole $a = 1$ $b = \sqrt{\frac{5}{3}}$

Asintoti $r_1 : y' = \frac{b}{a} x'$

$r_2 : y' = -\frac{b}{a} x'$

$\frac{b}{a} = \sqrt{\frac{5}{3}} > 1$



$\ast = \pm\sqrt{5}$

$\circ = \pm\sqrt{5}$

$$\begin{bmatrix} x \\ y \end{bmatrix} = P \begin{bmatrix} x' \\ y' \end{bmatrix} \Rightarrow \begin{bmatrix} x' \\ y' \end{bmatrix} = {}^tP \begin{bmatrix} x \\ y \end{bmatrix} \Longleftrightarrow \begin{cases} x' = \frac{1}{\sqrt{2}}(x + y) \\ y' = \frac{1}{\sqrt{2}}(-x + y) \end{cases}$$

$$y' = \frac{b}{a} x' \Longleftrightarrow \cancel{\frac{1}{\sqrt{2}}}(-x + y) = \frac{b}{a} \cancel{\frac{1}{\sqrt{2}}}(x + y)$$

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